

MERIT: A Sustainable DNN Accelerator Design with Photonic Phase-Change Memory

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Abstract—The growing computational demands of deep learning have driven interest in analog neural networks using resistive memory and silicon photonics. However, these technologies face inherent limitations in computing parallelism when used independently. Photonic phase-change memory (PCM), which integrates photonics with PCM, overcomes these constraints by enabling simultaneous processing of multiple inputs encoded on different wavelengths, significantly enhancing parallel computation for deep neural network (DNN) inference and training. This paper presents MERIT, a sustainable DNN accelerator that capitalizes on the non-volatility of resistive memory and the high operating speed of photonic devices. MERIT enables seamless inference and training by loading weight kernels into photonic PCM arrays and selectively supplying light encoded with input features for the forward pass and loss gradients for the backward pass. We compare MERIT with state-of-the-art digital and analog DNN accelerators including TPU, DEAP, and PTC. Simulation results demonstrate that MERIT reduces execution time by 68% and energy consumption by 64% for inference, and reduces execution time by 79% and energy consumption by 84% for training.

Index Terms—Silicon photonics, deep neural network, accelerator, inference, training

I. INTRODUCTION

THE exponentially growing computational demands of modern deep neural network (DNN) models call for scalable and efficient computing systems. Conventional digital systems struggle with high power consumption and limited scalability [1], making them insufficient for large-scale DNN applications. In contrast, analog systems have emerged as a promising alternative, leveraging their natural efficiency in performing dot product computation [2], [3], a core component of both DNN inference and training.

Previous analog computing systems have utilized memristor technology [4] to perform dot product computation for DNN inference and training [5], [6], [7], [8], [9], [10], [11], [12], [13], [14]. Memristor technology is known for its small device footprint and non-volatility, but it faces limitations in operating speed and scalability because of process variations and signal fluctuations [8]. Additionally, other analog computing systems leverage photonic devices [15], [16], [17], including Mach-Zehnder interferometers [18], [19], microdisks [20], [21], [22], and microring resonators (MRRs) [23], [24], [25], [26], [27], to perform similar computation. Silicon photonics technology

offers high bandwidth [28], [29], [30] and computation parallelism through wavelength-division multiplexing, and improved energy efficiency, facilitating the implementation of sustainable neural network architectures. However, it suffers from relatively larger device footprint and excess power consumption for the tuning process [15] to countermeasure process and temperature variations, albeit achieving high operating speed and scalability. Despite their advantages, both memristor-based and photonics-based analog computing systems share a common limitation: they only allow one single operation at a time on a basic device, significantly restricting achievable parallelism. For instance, a memristor device produces a single current output based on the applied input voltage and its conductance, while an MRR device generates a single photocurrent output by modulating the intensity of one resonant wavelength at a time.

Photonic phase-change memory (PCM) has emerged as a promising solution for implementing analog computing systems [31], [32], [33], combining the strengths of both memristor and silicon photonics technologies while overcoming their inherent limitations. A key advantage of photonic PCM devices is their ability to process multiple inputs encoded on different wavelengths simultaneously, enabling high computation parallelism without interference. This capability positions photonic PCM as a transformative technology for DNN inference and training, offering new opportunities for architecting sustainable neural network architectures.

In this paper, we introduce MERIT, a unified photonic DNN accelerator designed to leverage the unique benefits of photonic PCM for high computation parallelism and energy efficiency. MERIT features two main innovations: (1) achieving maximum parallelism by leveraging the one-step computation capability of photonic PCM devices and concurrently processing multiple wavelengths and (2) supporting DNN inference and training by fixing weight kernels in photonic PCM devices and selectively feeding vectorized input features and loss gradients to perform dot product computation. These innovations, combined with the inherent technological advantages of photonic PCM, enable MERIT to outperform state-of-the-art digital and analog DNN accelerators [5], [32], [34], [35]. Simulation results demonstrate that MERIT reduces execution time and energy consumption by 68% and 64% for inference, and by 79% and 84% for training. The major contributions of this paper are as follows:

- **Unified Photonic Architecture:** A versatile photonic architecture that supports both DNN inference and training by supplying vectors of input features and loss gradients to photonic PCM arrays with preloaded weight kernels.
- **Maximum Parallelism:** Enhanced parallelism through one-step dot product computation. During the forward pass,

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MERIT accumulates multiplication results from different input feature channels while sharing input features among different weight kernels. During the backward pass, it accumulates multiplication results from different output feature gradient channels while sharing loss gradients among different input feature channels. This is achieved by the optimized mapping of weight kernels on the photonic PCM arrays.

- *Performance Evaluation:* We evaluate MERIT and other state-of-the-art digital and analog DNN accelerators such as TPU [34], DEAP [35], Pipelayer [5], and PTC [32] using a collection of DNN models [36], [37], [38], [39], [40], [41]. Simulation results demonstrate the significant improvements in MERIT on execution time and energy consumption.

II. BACKGROUND & MOTIVATION

A. DNN Computation

A convolutional layer in a DNN model is typically described with eight indices, namely the number of input feature channels $\langle c \rangle$, the number of output feature channels $\langle k \rangle$, the height $\langle r \rangle$ and width $\langle s \rangle$ of the weight kernels, the height $\langle h \rangle$ and width $\langle w \rangle$ of the input feature channels, and the height $\langle e \rangle$ and width $\langle f \rangle$ of the output feature channels. Note the indices $\langle h \rangle$ and $\langle w \rangle$ can be derived by four indices $\langle r \rangle$, $\langle s \rangle$, $\langle e \rangle$, and $\langle f \rangle$, a convolutional layer in a DNN model can therefore be represented as a vector $\langle R, S, E, F, C, K \rangle$, wherein each vector element indicates the range of the respective index.

1) *DNN Inference:* DNN inference only includes forward propagation of the input features by sliding the weight kernels across the input feature channels and performing dot product computation between the vectorized weight kernels and input features in the respective sliding windows, as described in Equ. (1), assuming a stride of one. During the process, multiplication results $F_{rsc k} \times X_{r+e-1s+f-1c}$ are accumulated along the $\langle r \rangle$, $\langle s \rangle$, and $\langle c \rangle$ dimensions while input features $X_{r+e-1s+f-1c}$ are reused along the $\langle k \rangle$ dimension. Fig. 1 (a) shows the dot product computation in the forward pass of a convolutional layer $\langle R, S, E, F, C, K \rangle = \langle 2, 2, 2, 2, 1, 1 \rangle$. Output feature O_{11} is derived by performing dot product between a weight vector $\langle F_{11}, F_{12}, F_{21}, F_{22} \rangle$ and an input feature vector in respective sliding window $\langle X_{11}, X_{12}, X_{21}, X_{22} \rangle$. Other output features can be similarly derived and taken as the input features of the subsequent convolutional layer after being processed by a proper activation function.

$$O_{efk} = \sum_c \sum_s \sum_r F_{rsc k} \times X_{r+e-1s+f-1c} \quad (1)$$

2) *DNN Training:* DNN training includes both backward propagation of loss gradients and weight update, in addition to the forward propagation of input features, which is the same as in DNN inference. Backward propagation of loss gradients [42] is achieved by sliding the weight kernels across the output feature gradient channels and performing dot product computation between the vectorized weight kernels and output feature gradients in the respective sliding windows, as described in Equ. (2), assuming a stride of one. During this

process, multiplication results $F_{rsc k} \times \partial L / \partial O_{h-r+1w-s+1k}$ are accumulated along the $\langle r \rangle$, $\langle s \rangle$, and $\langle k \rangle$ dimensions while output feature gradients $\partial L / \partial O_{h-r+1w-s+1k}$ are reused along the $\langle c \rangle$ dimension. Note that both the input and output feature gradient channels are vertically and horizontally flipped due to the negative signs of r and s in the output feature gradient term in Equ. (2). Additionally, the output feature gradient channels are zero-padded to ensure the proper size of the derived input feature gradient channels. Fig. 1 (b) illustrates the dot product computation in the backward pass of the convolutional layer $\langle R, S, E, F, C, K \rangle = \langle 2, 2, 2, 2, 1, 1 \rangle$. Input feature gradient $\partial L / \partial X_{33}$ is derived by performing dot product between the same weight vector as in the previous DNN inference example $\langle F_{11}, F_{12}, F_{21}, F_{22} \rangle$ and an output feature gradient vector in respective sliding window $\langle 0, 0, 0, \partial L / \partial O_{22} \rangle$. The rest input feature gradients can be similarly derived and taken as the output feature gradients of the prior convolutional layer after the Hadamard product with the derivative of the activation of the output features from the prior convolutional layer.

$$\frac{\partial L}{\partial X_{hwc}} = \sum_k \sum_s \sum_r F_{rsc k} \times \frac{\partial L}{\partial O_{h-r+1w-s+1k}} \quad (2)$$

Weight update is achieved by sliding the output feature gradient channels across the input feature channels and performing dot product computation between the vectorized output feature gradients and input features in the respective sliding windows, as described in Equ. (3), assuming a stride of one. During this process, multiplication results $\partial L / \partial O_{efk} \times X_{r+e-1s+f-1c}$ are accumulated along the $\langle e \rangle$ and $\langle f \rangle$ dimensions. Fig. 1 (c) illustrates the dot product computation in the weight update of the convolutional layer $\langle R, S, E, F, C, K \rangle = \langle 2, 2, 2, 2, 1, 1 \rangle$. Weight gradient $\partial L / \partial F_{11}$ is derived by performing dot product between $\langle \partial L / \partial O_{11}, \partial L / \partial O_{12}, \partial L / \partial O_{21}, \partial L / \partial O_{22} \rangle$, which is an output feature gradient vector, and an input feature vector in respective sliding window $\langle X_{11}, X_{12}, X_{21}, X_{22} \rangle$. The rest weight gradients can be similarly derived and utilized to update respective local weights. This work focuses on the acceleration of the forward propagation of input features and the backward propagation of loss gradients, by optimally mapping weight kernels on the photonic PCM arrays and selectively supplying vectorized input features and output feature gradients, respectively.

$$\frac{\partial L}{\partial F_{rsc k}} = \sum_f \sum_e \frac{\partial L}{\partial O_{efk}} \times X_{r+e-1s+f-1c} \quad (3)$$

B. Analog Dot Product Computation

1) *Memristor Device:* Memristor technology [4] has been utilized to architect analog computing systems for one-step dot product computation [2]. Fig. 2 (a) provides one example wherein each of the three memristor devices is used to connect a respective horizontal wire and the vertical wire and positioned at their intersection. These memristor devices are programmed to conductances G_1 , G_2 , and G_3 to represent a vector $\mathbb{G}$. In addition, voltage levels V_1 , V_2 , and V_3 representing another vector $\mathbb{V}$ are supplied to the respective horizontal wires. Based

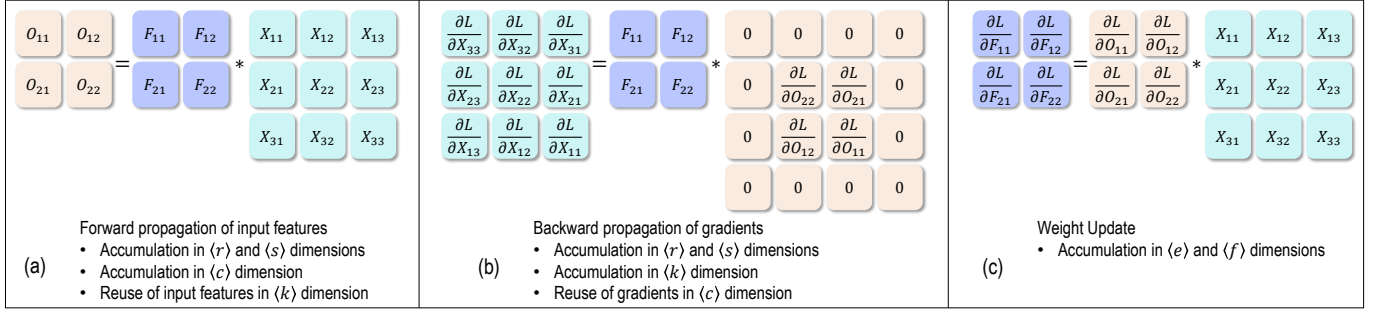


Fig. 1. Dot product computation for (a) forward propagation of input features, (b) backward propagation of loss gradients, and (c) weight update of a sample convolutional layer $\langle R, S, E, F, C, K \rangle = \langle 2, 2, 2, 2, 1, 1 \rangle$.

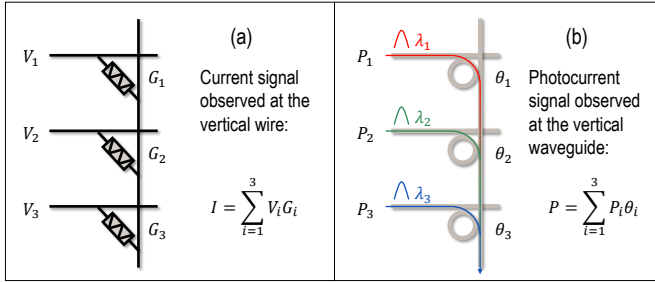


Fig. 2. Dot product computation on (a) memristor devices and (b) microring resonators with respective resonant wavelengths.

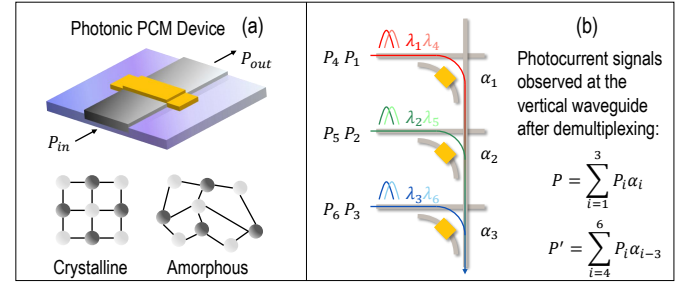


Fig. 3. Architectures of (a) a photonic phase-change memory device and (b) a crossbar for dot product computation.

on Kirchhoff's Law, one current signal $I = V_1 \times G_1 + V_2 \times G_2 + V_3 \times G_3$ can be observed and measured at the end of the vertical wire, representing the dot product of vectors $\mathbb{V}$ and $\mathbb{G}$. Such architectures can potentially achieve high computation density and energy efficiency due to small footprint and non-volatility properties of memristor devices. However, they struggle with limitations in operating speed and scalability due to process variations and signal fluctuations [8]. Moreover, the dot product between only one vector $\mathbb{V}$ and the programmed vector $\mathbb{G}$ can be performed at any given time, which significantly restricts achievable computation parallelism.

2) *Photonic Device*: Silicon photonics technology such as MRR [43] has also been utilized to architect analog computing systems for one-step dot product computation [16], [17]. Fig. 2 (b) provides an example wherein each of the three MRRs is located at the intersection of a respective horizontal waveguide and the vertical waveguide and couples a resonant wavelength from the horizontal waveguide to the vertical waveguide after proper attenuation. The attenuation factors for these MRRs are programmed to θ_1, θ_1 , and θ_3 to represent a vector $\mathbb{A}$. In addition, wavelengths λ_1, λ_2 , and λ_3 with respective intensities P_1, P_2 , and P_3 representing another vector $\mathbb{P}$ are supplied to the respective horizontal waveguides. The intensity of combined wavelengths $P = P_1 \times \theta_1 + P_2 \times \theta_2 + P_3 \times \theta_3$ can be sensed and measured by a photodetector at the end of the vertical waveguide, representing the dot product of vectors $\mathbb{P}$ and $\mathbb{A}$. Such architectures can potentially achieve high operating speed and scalability due to high modulation speed and low

insertion loss. However, they struggle with the relatively large footprint of photonic devices and consume excess power for electrical or thermal tuning to compensate for the process and temperature variations and maintain programmed attenuation factors. Moreover, the dot product between only one vector $\mathbb{P}$ and the programmed vector $\mathbb{A}$ can be performed at any given time, similarly restricting achievable computation parallelism.

3) *Photonic Phase-Change Memory Device*: Nascent photonic PCM technology [31], [44], [45] has recently been used to architect analog computing systems for one-step dot production computation [32], [33], [46], [47]. One essential characteristic of photonic PCM devices is their high contrast in both electrical (resistivity) and optical (refractive index) properties between amorphous and crystalline states. Consider a photonic PCM device positioned above a waveguide as shown in Fig. 3 (a), its state can be either read through measuring the attenuation of light intensity $P_{out} - P_{in}$, or programmed by supplying picosecond optical pulses [32]. This photonic PCM device represents 0 if programmed to the crystalline state when most of the light is absorbed and represents 1 if programmed to the amorphous state when close to none of the light is absorbed. Moreover, it represents a specific number between 0 and 1 if programmed to a respective transient state between crystalline and amorphous states when the light is partially absorbed. The dispersion in attenuation factor can be neglected in the adopted 180 – 205 THz wavelength spectrum, or be calibrated through adjusting wavelength intensities [32]. The reliability of adopted photonic PCM devices for one-step dot product computation

has been validated in a 600-hour durability test that showcases the resilience under continuous operation [32]. Write cycling endurance of 10^{12} has been reported and 10^{15} is expected [31].

Fig. 3 (b) provides an example wherein each of the three photonic PCM devices is coupled with a respective horizontal waveguide and the vertical waveguide via directional couplers [48] and positioned at their intersection. These photonic PCM devices are programmed to transient state corresponding to attenuation factors α_1 , α_2 , and α_3 to represent a vector $\mathbb{A}$. In addition, wavelengths λ_1 , λ_2 , and λ_3 with respective intensities P_1 , P_2 , and P_3 representing another vector $\mathbb{P}_1$ are supplied to the respective horizontal waveguides. The intensity of combined wavelengths $P = P_1 \times \alpha_1 + P_2 \times \alpha_2 + P_3 \times \alpha_3$ can be sensed and measured by a photodetector at the end of the vertical waveguide, representing the dot product of vectors $\mathbb{P}_1$ and $\mathbb{A}$. Such architectures can leverage the high computation density and energy efficiency similar to the advantages of memristor technology, as well as the high operating speed and scalability similar to the advantages of silicon photonics. More importantly, as shown in Fig. 3 (b), since the dispersion in attenuation factor can be neglected in the adopted wavelength spectrum, it is possible to supply another three wavelengths λ_4 , λ_5 , and λ_6 with respective intensities P_4 , P_5 , and P_6 representing a third vector $\mathbb{P}_2$ to the respective horizontal waveguides. The intensity of combined wavelengths $P' = P_4 \times \alpha_1 + P_5 \times \alpha_2 + P_6 \times \alpha_3$ can be sensed and measured by another photodetector at the end of the vertical waveguide after proper wavelength demultiplexing, representing the dot product of vectors $\mathbb{P}_2$ and $\mathbb{A}$. We leverage the additional parallelism in the frequency (wavelength) domain to architect a unified architecture for inference and training by optimally mapping weight kernels to photonic PCM arrays and selectively supplying vectorized input features and output feature gradients in MERIT.

III. MERIT ARCHITECTURE

A. Key Building Devices

1) *Directional Coupler*: Fig. 4 (a) shows the architecture of a directional coupler [48] used in the MERIT design. This coupler comprises a symmetric coupler, an asymmetric-waveguide-based phase control section, and a second symmetric coupler, connected by linearly tapered waveguides. The lengths of the symmetric couplers and the phase control section are denoted as L_1 and L_2 , respectively. E_1 and E_2 represent the electric fields at the respective input ports while E_3 and E_4 represent the electric fields at the respective output ports. The split ratio of the cross port relative to the input port depends on parameters such as L_1 and L_2 . In the MERIT design, a directional coupler with a maximum coupling length of $10 \mu\text{m}$ supports all required split ratios [32]. These directional couplers are critical for evenly distributing light among different columns of photonic PCM devices in a photonic PCM array. As passive components, these directional couplers consume no power during operation, aside from insertion loss.

2) *Tunable Microring Resonator*: Unlike MRRs that operate strictly at on-resonance and off-resonance states for optical modulation and filtering, tunable MRR functions in the transient state between these two extremes [49], [50], [51]. As shown in

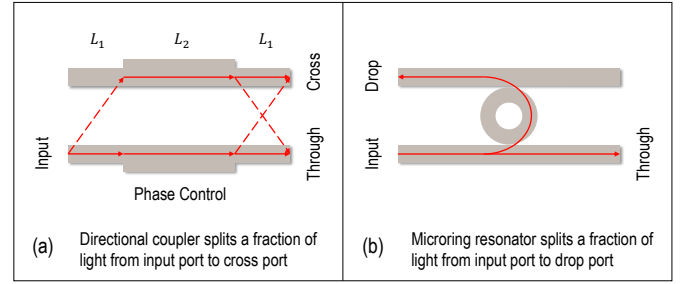


Fig. 4. Architectures and working mechanisms of (a) directional coupler and (b) tunable microring resonator.

Fig. 4 (b), the interior and exterior regions of a tunable MRR are doped to form a PIN diode structure. In the absence of a bias voltage, the tunable MRR remains inactive (off-resonance), allowing light to pass uninterrupted from the input port to the through port. When a bias voltage is applied to the PIN diode, the MRR splits incoming light at the input port, directing α fraction to the drop port while transmitting the remainder $1 - \alpha$ to the through port. This configuration results in an attenuation factor of $1 - \alpha$. The bias voltage applied to the PIN diode is controlled by a digital-to-analog converter (DAC), enabling precise tuning of the attenuation factor within a specified range. In the MERIT design, tunable MRRs are crucial for encoding input feature or output feature gradient vectors.

B. MERIT Architecture

Fig. 5 shows a sample MERIT tile architecture with a 4×4 photonic PCM array for dot product computation. The MERIT tile consists of five critical components: wavelength generation, input vector generation, photonic PCM array for dot product computation, output vector collection, and SRAM buffer. Inside the wavelength generation component, an off-chip laser source is connected to a microcomb [52], [53] to generate required wavelengths, each routed through a DEMUX to a respective waveguide. Equally spaced wavelengths are then merged into a composite waveguide through a MUX, which is coupled to the respective horizontal waveguide inside the input vector generation component and then to the photonic PCM array. Multiple such composite waveguides are created by merging different sets of wavelengths through additional MUXes. Inside the input vector generation component, wavelengths carried on each horizontal waveguide are encoded as elements at the corresponding positions of different input vectors through tunable MRRs controlled by DACs. The encoded input vectors are then supplied to the photonic PCM array for dot product computation. The photonic PCM array comprises photonic PCM devices positioned at the intersections of horizontal and vertical waveguides. The attenuation factors of photonic PCM devices in each array column are programmed to represent a specific vector. Hence, different columns in the array represent different vectors. Dot product computation is carried out in parallel between each pair of an encoded input vector and a vector programmed on the photonic PCM array. Wavelengths carried by each vertical waveguide of the photonic PCM array

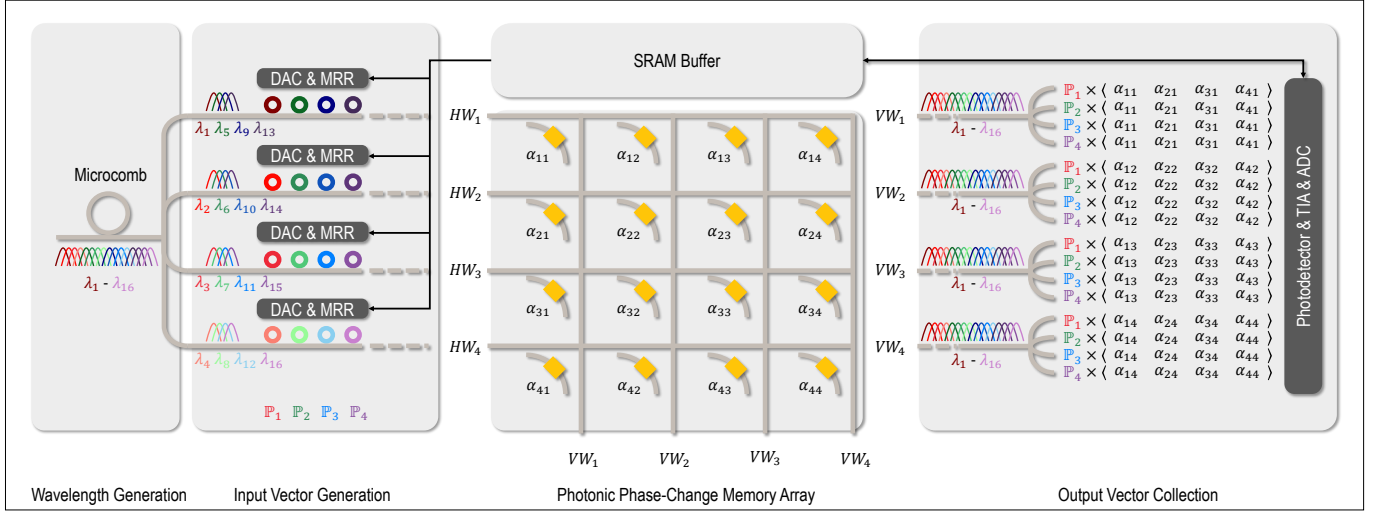


Fig. 5. MERIT tile architecture with a 4×4 photonic PCM array and sixteen available wavelengths from λ_1 to λ_{16} for dot product computation. This MERIT tile supports concurrent dot product computation between each of the four encoded vectors $\mathbb{P}_1, \mathbb{P}_2, \mathbb{P}_3$, and $\mathbb{P}_4$ and each of the four vectors programmed on the respective columns of the photonic PCM array, $\langle \alpha_{11}, \alpha_{21}, \alpha_{31}, \alpha_{41} \rangle$, $\langle \alpha_{12}, \alpha_{22}, \alpha_{32}, \alpha_{42} \rangle$, $\langle \alpha_{13}, \alpha_{23}, \alpha_{33}, \alpha_{43} \rangle$, and $\langle \alpha_{14}, \alpha_{24}, \alpha_{34}, \alpha_{44} \rangle$.

are routed through a DEMUX to the respective waveguides while different sets of adjacent wavelengths are then merged into composite waveguides through MUXes in the output vector collection component. A photodetector at the end of each such composite waveguide senses and measures a distinct dot product result. SRAM buffer stores intermediate data to ensure seamless data flow within a MERIT tile. We explain the collaborative functioning of these five components in detail using the example provided in Fig. 5.

1) *Wavelength Generation*: In the example, shown in Fig. 5, an off-chip laser source is connected to a microcomb to generate sixteen wavelengths ranging from λ_1 to λ_{16} , all carried on a single waveguide. This waveguide is split into sixteen separate waveguides, each carrying a corresponding wavelength, through a DEMUX. Waveguides carrying a set of four equally spaced wavelengths (e.g., $\lambda_1, \lambda_5, \lambda_9$, and λ_{13}) are then merged into a composite waveguide through a MUX and coupled with a respective horizontal waveguide in the input vector generation component. There are four such composite waveguides carrying four sets of wavelengths.

2) *Input Vector Generation*: In the example, shown in Fig. 5, the input vector generation component includes four horizontal waveguides, each carrying a set of four wavelengths (e.g., $\lambda_1, \lambda_5, \lambda_9$, and λ_{13} on the top horizontal waveguide). Hence, the component can concurrently generate four four-element vectors $\mathbb{P}_1, \mathbb{P}_2, \mathbb{P}_3$, and $\mathbb{P}_4$. Specifically, elements from vector $\mathbb{P}_1$ are encoded on wavelengths $\lambda_1, \lambda_2, \lambda_3$, and λ_4 using respective tunable MRRs and DACs. Elements from vectors $\mathbb{P}_2, \mathbb{P}_3$, and $\mathbb{P}_4$ are encoded on subsequent wavelengths. We utilize adjacent wavelengths to encode input vectors for convenient collection of dot product results. One consequence of such an arrangement is that each horizontal waveguide inside the component carries equally spaced wavelengths and contributes to the generation of a specific element from every input vector.

3) *Photonic PCM Array*: The MERIT tile illustrated in Fig. 5 features a 4×4 photonic PCM array, where each photonic PCM device is positioned at an intersection of the horizontal and vertical waveguides. Each photonic PCM device precisely couples a portion of light from the corresponding horizontal wavelength via a directional coupler and then fully couples the attenuated light back into the respective vertical waveguide via a second directional coupler. Each photonic PCM device is programmed to a specific attenuation factor, representing an element of an encoded vector. Within each row of the array, all photonic PCM devices receive an equal portion of light from the same horizontal waveguide, which is achieved by tuning the split ratio (coupling distance) of each relevant directional coupler at fabrication time. For instance, the four directional couplers coupling the top horizontal waveguide HW_1 to four photonic PCM devices with attenuation factors $\alpha_{11}, \alpha_{12}, \alpha_{13}$, and α_{14} are tuned to have split ratios of $1/4, 1/3, 1/2$, and $1/1$, respectively. These split ratios apply to all four wavelengths $\lambda_1, \lambda_5, \lambda_9$, and λ_{13} carried on waveguide HW_1 . A similar arrangement can be inferred in the rest rows of the array. On the other hand, within each column of the array, all photonic PCM devices completely couple the attenuated light back into the same vertical waveguide, which is also achieved by tuning the split ratio of each relevant directional coupler at fabrication time. For instance, the four directional couplers coupling the left vertical waveguide VW_1 to four photonic PCM devices with attenuation factors $\alpha_{11}, \alpha_{21}, \alpha_{31}$, and α_{41} are all tuned to have the same split ratio of $1/1$. A similar arrangement can also be inferred in the rest columns of the array. Four vectors $\mathbb{A}_1, \mathbb{A}_2, \mathbb{A}_3$, and $\mathbb{A}_4$ are programmed on the array, and the dot product computation results between one of them and all four encoded input vectors can be sensed and measured at the end of a respective vertical waveguide. For instance, the dot product computation results between the programmed vector $\mathbb{A}_1$ and four encoded input vectors $\mathbb{P}_1, \mathbb{P}_2, \mathbb{P}_3$, and $\mathbb{P}_4$ can

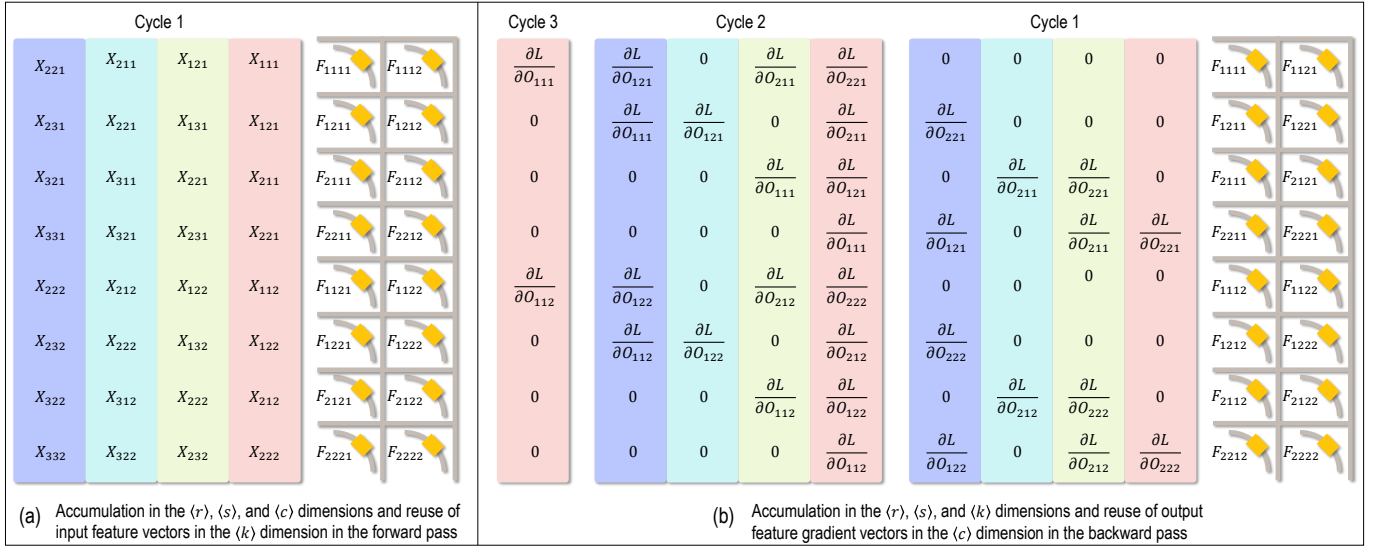


Fig. 6. Dot product computation during (a) forward propagation of input features and (b) backward propagation of output feature gradients for a sample convolutional layer $\langle R, S, E, F, C, K \rangle = \langle 2, 2, 2, 2, 2, 2 \rangle$ on an 8×2 photonic PCM array, by optimally mapping weight kernels on the photonic PCM array and selectively supplying vectorized input features and output feature gradients, respectively.

be sensed and measured by photodetectors at the end of the vertical waveguide VW_1 .

4) *Output Vector Collection*: In the output vector collection component, sixteen wavelengths carried by each of the vertical waveguides from the photonic PCM array are split and then merged into four sets, each including four adjacent wavelengths. For instance, the vertical waveguide VW_1 is split into sixteen separate waveguides through a DEMUX, each carrying a certain wavelength, and then merged into four composite waveguides through MUXes, carrying separate sets of adjacent wavelengths λ_1 to λ_4 , λ_5 to λ_8 , λ_9 to λ_{12} , and λ_{13} to λ_{16} . A photodetector at the end of each composite waveguide is utilized to sense and measure the dot product computation result between a pair of a vector programmed on the photonic PCM array and an encoded input vector. Photocurrent signals are amplified by transimpedance amplifiers (TIAs) and converted to the digital format through analog-to-digital converters (ADCs).

5) *SRAM On-Chip Buffer*: The SRAM stores intermediate data for dot product computation. During forward propagation, it holds input features fetched from off-chip memory, partial sums, and output features. During backward propagation, it also holds loss gradients.

6) *System Scalability*: MERIT tile adopts 64×64 photonic PCM array, forecast by recent photonic PCM accelerators for DNN applications [32]. MERIT architecture includes multiple MERIT tiles, which are optically connected to the laser source using space-division multiplexing for dot product computation, and electrically connected to each other and off-chip memory via associated routers [54] for data exchange. Four factors are considered when determining the optimal size of the photonic PCM array, namely, laser power, quantization loss, DAC/ADC overhead, and data movement associated with tiling. As the size of the photonic PCM array grows, the accumulated insertion loss increases linearly, leading to an exponential rise in required

laser power [49]. Meanwhile, as the array size increases, each photodetector and associated ADC must measure the intensity of light carrying more multiplication results. Quantization is applied to prevent overflow, however, compromising accuracy of accumulation results and adversely affecting the performance of DNN inference and training. Furthermore, increase in array size facilitates the accommodation of larger processing tiles, minimizing associated data movement [51] and DAC/ADC overhead during transitions between digital and analog domains. The chosen 64×64 array size results from a tradeoff: while increasing the array size reduces data movement and DAC/ADC overhead, it also leads to higher laser power consumption and quantization loss. The optimal array size may vary depending on factors such as the insertion loss of photonic components, the resolution of DACs, ADCs, and photonic PCM devices, and the characteristics of the DNN applications being processed.

C. Uniform Architecture for DNN Inference and Training

We explain how the proposed MERIT architecture supports both DNN inference and training by optimally mapping weight kernels on the photonic PCM array and selectively supplying vectorized input features and output feature gradients during forward and backward passes, respectively, with a sample convolutional layer $\langle R, S, E, F, C, K \rangle = \langle 2, 2, 2, 2, 2, 2 \rangle$ executed on an 8×2 photonic PCM array.

1) *Forward Propagation*: As depicted in Fig. 6 (a), during the forward propagation of input features, the first weight kernel $\langle F_{1111}, \dots, F_{2211}, F_{1121}, \dots, F_{2221} \rangle$ is mapped to the first column of the photonic PCM array. The second weight kernel $\langle F_{1112}, \dots, F_{2212}, F_{1122}, \dots, F_{2222} \rangle$ is similarly mapped to the second column. Such a mapping strategy ensures one-step dot product computation between weight kernels and vectorized input features and seamless sharing of input features between different weight kernels, aligning with the analysis in Fig. 1

(a). For instance, vector including input features from the first sliding window $\langle X_{111}, \dots, X_{221}, X_{112}, \dots, X_{222} \rangle$ is carried by a first set of eight wavelengths λ_1 to λ_8 and supplied to both columns of the photonic PCM array for concurrent dot product computation with two mapped weight kernels. Similarly, vectors including input features from the subsequent sliding windows are carried by another three sets of eight wavelengths λ_9 to λ_{16} , λ_{17} to λ_{24} , and λ_{25} to λ_{32} and supplied to both columns of the photonic PCM array. As photonic PCM devices possess the capability of attenuating multiple wavelengths in parallel, all four input feature vectors are processed in the same cycle, maximizing computation parallelism. The number of required wavelengths is determined by the size of the photonic PCM array as well as the number of supplied input vectors that can be processed in parallel.

2) *Backward Propagation*: As shown in Fig. 6 (b), during backward propagation of output feature gradients, first weight kernel $\langle F_{1111}, \dots, F_{2211}, F_{1112}, \dots, F_{2212} \rangle$ is mapped to the first column of the photonic PCM array while the second weight kernel $\langle F_{1121}, \dots, F_{2221}, F_{1122}, \dots, F_{2222} \rangle$ is mapped to the second column. Please note that, unlike in forward propagation of input features where weights from different input channels of the same weight kernel are mapped to the same array column for one-step accumulation of multiplication results in the $\langle r \rangle$, $\langle s \rangle$, and $\langle c \rangle$ dimensions, in backward propagation of output feature gradients, here weights from the same input channel of different weight kernels are mapped to the same column for one-step accumulation of multiplication results in the $\langle r \rangle$, $\langle s \rangle$, and $\langle k \rangle$ dimensions. Such a mapping strategy ensures one-step dot product computation between weight kernels and vectorized output feature gradients and seamless sharing of output feature gradients between different weight input channels, which aligns with the analysis in Fig. 1 (b). For instance, the vector including output feature gradients and padded zeros from the first sliding window $\langle 0, 0, 0, \partial L / \partial O_{221}, 0, 0, 0, \partial L / \partial O_{222} \rangle$ is carried by the same set of eight wavelengths λ_1 to λ_8 and supplied to both columns of photonic PCM array for concurrent dot product computation with two mapped weight kernels. Similarly, vectors including input features from the subsequent sliding windows are carried by the following three sets of eight wavelengths λ_9 to λ_{16} , λ_{17} to λ_{24} , and λ_{25} to λ_{32} and supplied to both columns of the photonic PCM array. Four output feature gradient vectors can be processed in the same cycle for maximum computation parallelism. As each output feature gradient channel is padded with zeros to the size of 4×4 , there are nine output feature gradient vectors to be processed, corresponding to nine different sliding windows. Processing all nine output feature gradient vectors requires three cycles, as shown in Fig. 6 (b).

IV. EVALUATION METHODOLOGY

A. Evaluation Setup

We have extended the open-source ASTRA-Sim simulator to model MERIT and other baseline architectures for both DNN inference and training applications. Some key parameters used for modeling are listed in Table I. Specifically, we consider 8-bit DAC and ADC with over 10 GS/s sampling speed (14 GS/s for DAC [55] and 10 GS/s for ADC [56]) as their

TABLE I
MERIT DEVICE PARAMETERS

DEVICE	PARAMETER	VALUE
DAC	Power	50 mW [55]
	Area	11,000 μm^2 [55]
ADC	Power	15 mW [56]
	Area	2,850 μm^2 [56]
TIA	Power	3 mW [57]
	Area	50 μm^2 [57]
MRR	Power	0.2 mW [58]
	Area	93 μm^2 [59]
	Insertion Loss	0.01 dB [49]
Photodetector	Power	1.1 mW [60]
	Area	40 μm^2 [60]
	Sensitivity	-23.4 dBm [61]
Directional Coupler	Insertion Loss	0.1 dB [32]
	Coupling Length	10 μm [32]
Laser Source	Wall-Plug Efficiency	0.2 [58]
Photonic PCM Array	Array Size	64×64
	Device Size	$30\mu\text{m} \times 30\mu\text{m}$ [32]
	Tuning Pulse	200 ns [32]
SRAM Buffer	Capacity	32 MiB
DRAM	Capacity	32 GiB
	Bandwidth	1200 GB/s

sampling speed limits the achievable operating speed of MERIT architecture. Prototype of photonic PCM device achieves 5-bit resolution [32] while the 8-bit resolution is projected [21], [62]. Sampling speed beyond 10 GS/s entails elevated power consumption and area requirement. MERIT and other baseline architectures are equipped with 32 MiB SRAM buffer and 32 GiB HBM DRAM with 1200 GB/s bandwidth [34] for a fair comparison. The computation load for TPU [34] is calculated by the involved SCALE-Sim simulator specific to systolic array architecture while the computation loads for MERIT and other analog baseline architectures are derived based on the overall numbers of operations, operating speed, and device count.

B. Power Model

We adopt the following power model to calculate the required laser power. The laser power per wavelength is obtained using Equ. (4), where P_{rs} represents the sensitivity of photodetectors (-23.4 dBm [61]) and system margin M_{system} (4 dB [49]) accounts for possible power penalties developed during system lifetime. The overall insertion loss C_{loss} can be obtained by accumulating the insertion loss of the traversed photonic components. Power consumption of the off-chip laser source is derived based on laser power per wavelength P_{laser} , wavelength count, and laser source wall-plug efficiency.

$$P_{laser} = P_{rs} + C_{loss} + M_{system} \quad (4)$$

C. Baseline Architectures

MERIT architecture is compared against other state-of-the-art digital and analog accelerator architectures for DNN inference

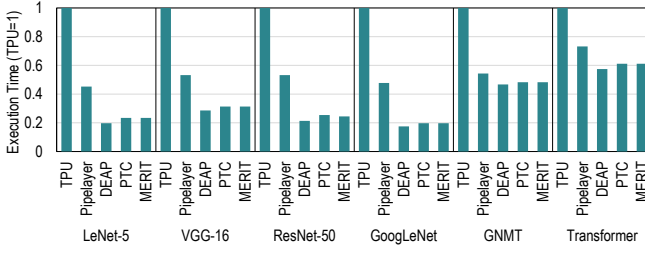


Fig. 7. Execution time comparison between MERIT and other four architectures (normalized to TPU) for DNN inference.

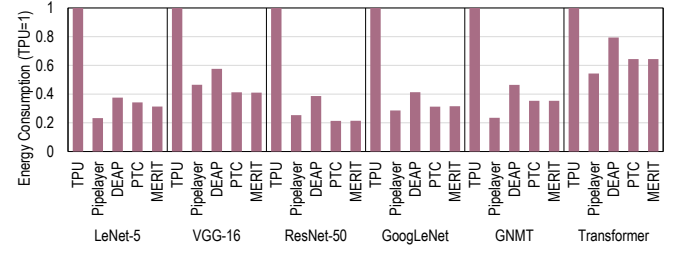


Fig. 10. Energy comparison between MERIT and other four architectures (normalized to TPU) for DNN inference.

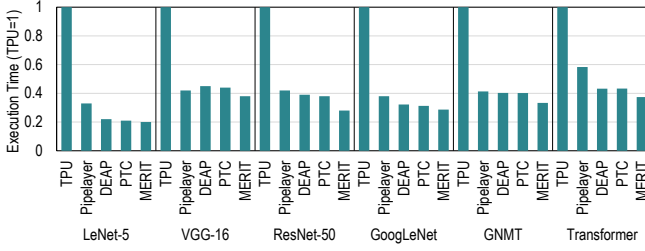


Fig. 8. Execution time comparison between MERIT and other four architectures (normalized to TPU) for the stochastic gradient descent training.

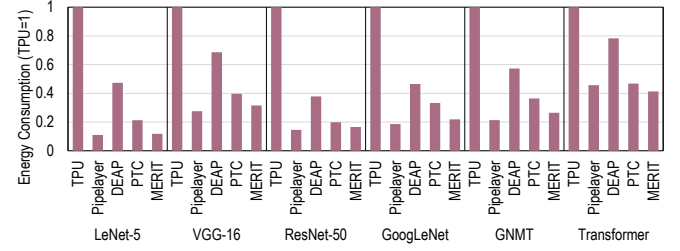


Fig. 11. Energy comparison between MERIT and other four architectures (normalized to TPU) for the stochastic gradient descent training.

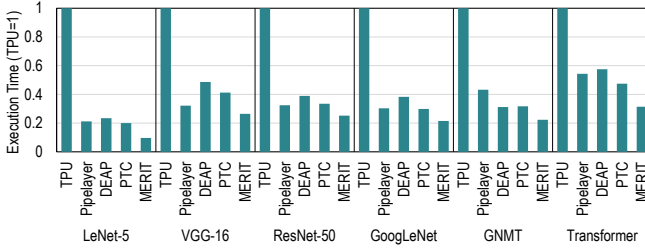


Fig. 9. Execution time comparison between MERIT and other four architectures (normalized to TPU) for the mini-batch (128 size) gradient descent training.

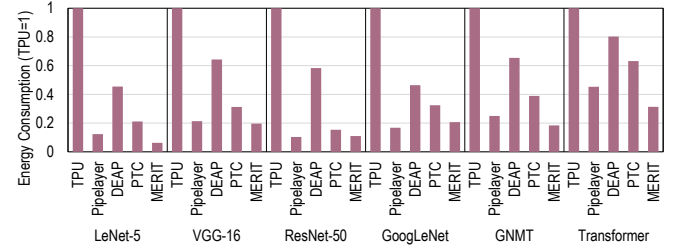


Fig. 12. Energy comparison between MERIT and other four architectures (normalized to TPU) for the mini-batch (128 size) gradient descent training.

and training applications. Baseline TPU [34] is a digital systolic-array-like accelerator architecture indicating digital computing systems' achievable performance and energy efficiency. Another baseline Pipelayer [5] is an analog memristor-based electrical accelerator architecture while baseline DEAP [35] is an analog MRR-based optical photonic accelerator architecture. Pipelayer and DEAP leverage the small device footprint and non-volatility advantages of memristor technology and high operating speed and scalability advantages of silicon photonics, respectively. Baseline PTC [32] is an analog photonic PCM-based accelerator architecture, which exploits the same photonic PCM devices as MERIT. The key improvement of MERIT over PTC is the capability of concurrently processing the forward propagation of input features and backward propagation of output feature gradients in DNN training.

D. Evaluation Benchmark

We adopt six DNN models for evaluation: LeNet-5 [36] with the MNIST dataset for digit recognition, VGG-16 [37], ResNet-50 [39], and GoogLeNet [38] with the ImageNet dataset for image classification, and Google Neural Machine Translation (GNMT) [40], [41], [63] with the WMT16 EN-DE dataset as

well as Transformer [41], [64] with the WMT17 EN-DE dataset for language translation. These adopted DNN models enable a comprehensive evaluation of MERIT and all the baseline architectures as they cover a wide range of applications and vary in model architectures. Specifically, LeNet-5 consists of three convolutional layers interleaved with two pooling layers, followed by two fully connected layers with around 60 thousand parameters. VGG-16 includes thirteen convolutional layers and three fully connected layers, organized into five convolutional blocks, each followed by a pooling layer with about 138 million parameters. ResNet-50 has fifty layers including convolutional layers, batch normalization layers, activations, and identity or convolutional shortcut connections with 25 million parameters. GoogLeNet has twenty-two layers made up of several inception modules where multiple convolutions of different kernel sizes are performed in parallel alongside pooling operations with about 6.8 million parameters. GNMT has an encoder-decoder architecture with attention mechanisms where both encoder and decoder comprise eight recurrent layers with altogether 1 billion parameters. Transformer replaces the recurrent layers with self-attention mechanisms and has 213 million parameters.

V. EXPERIMENT RESULTS

A. Execution Time

1) *DNN Inference*: Fig. 7 demonstrates the execution time comparison between MERIT and other baseline architectures for inference of six DNN models. All the results are normalized to that of the TPU baseline architecture. We observe significant execution time reduction by comparing TPU with other analog computing architectures ranging from 26% to 82%, proving efficient one-step dot product computation advantage of analog computing architectures over digital counterparts. Compared with Pipelayer, DEAP gets execution time reduction ranging from 13% to 63%, proving the high operating speed advantage of photonic computing architectures over electrical memristor-based computing architectures. We assume the same operating speed (10 GHz) and peak performance (82 TOPS) for both DEAP and PTC. In real application scenarios, PTC achieves higher execution time ranging from 3% to 19% compared to DEAP, mainly because of the costly operations of programming photonic PCM devices. We will explore dataflow optimizations in our future work to minimize such programming operations. MERIT and PTC achieve similar performance as they function in essentially the same way.

2) *DNN Training*: Fig. 8 demonstrates the execution time comparison between MERIT and other baseline architectures for stochastic gradient descent training of six DNN models. All values are normalized to that of the TPU baseline architecture. We still observe significant execution time reduction ranging from 42% to 80% when comparing TPU with all other analog computing architectures, since both the forward propagation of input features and backward propagation of output feature gradients incur intensive dot product computation which can be adequately handled in the analog domain. Compared with Pipelayer, DEAP achieves execution time reduction ranging from 2% to 33% despite Pipelayer can duplicate and reorganize weight kernels for forward and backward passes to minimize data movement. PTC takes slightly shorter execution time than DEAP ranging from 2% to 5% as the overhead induced by the costly operations of programming photonic PCM devices is completely offset by reusing weight kernels between forward and backward passes. Compared with PTC, MERIT achieves execution time reduction ranging from 5% to 26% as MERIT minimizes programming operations on photonic PCM devices.

Fig. 9 shows the execution time comparison between MERIT and other baseline architectures for mini-batch gradient descent training of six DNN models, assuming each mini-batch includes 128 examples. All values are normalized to that of the TPU baseline architecture. The execution time reduction of analog computing architectures compared to TPU further increases to the range of 42% to 90%, owing to the computation parallelism between training examples in the mini-batch gradient descent scenario. Notably, compared with PTC, MERIT gets execution time reduction ranging from 25% to 52%, which is significantly higher than in stochastic gradient descent training, since the programmed weight kernels on photonic PCM devices have been extensively reused between forward and backward passes of different training examples in a mini-batch, which is not possible in stochastic gradient descent training.

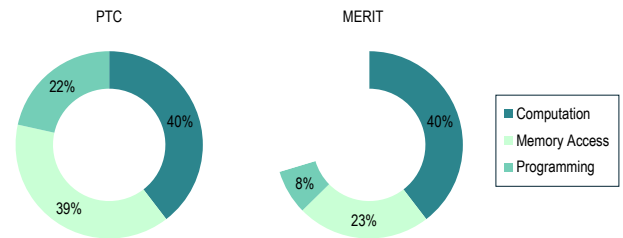


Fig. 13. Execution time breakdown of PTC baseline and MERIT architectures for mini-batch gradient descent training of GNMT model (normalized to PTC).

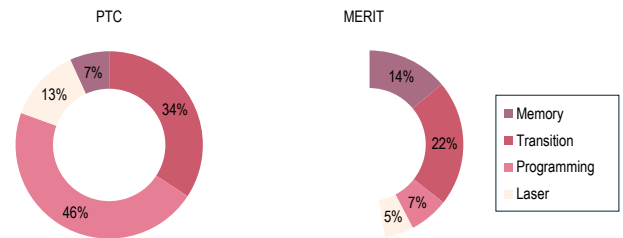


Fig. 14. Energy consumption breakdown of PTC and MERIT architectures for mini-batch gradient descent training of GNMT model (normalized to PTC).

B. Energy Consumption

1) *DNN Inference*: Fig. 10 shows the energy consumption comparison between MERIT and other baseline architectures for inference of six DNN models. All the values are normalized to that of the TPU baseline architecture. Significant decrease in energy consumption ranging from 21% to 79% is observed when comparing TPU with all other analog computing architectures, even for DEAP where excess energy is consumed to countermeasure process and temperature variations. We focus on the comparison of three photonic computing architectures, MRR-based architecture DEAP, photonic PCM-based architecture PTC, and photonic-based architecture with optimized weight mapping MERIT. Both PTC and MERIT consume 9% to 45% less energy compared with DEAP, proving the superior energy efficiency of photonic PCM devices due to their non-volatility characteristic. MERIT and PTC consume comparable energy since they are built upon the same photonic PCM devices and function in essentially the same way in DNN inference.

2) *DNN Training*: Fig. 11 illustrates the energy consumption comparison between MERIT and other baseline architectures for stochastic gradient descent training of six DNN models. All values are normalized to that of the TPU baseline architecture. All evaluated analog computing architectures achieve notable reduction in energy consumption compared with TPU, in range of 22% to 89%. Within four analog computing architectures, Pipelayer consumes the least energy as it fits more memristor arrays in a fixed chip area due to smaller footprint of memristor devices, which enables duplication and reorganization of weight kernels in forward and backward passes and minimizing data movement. DEAP consumes the highest energy mainly due to the necessity of tuning MRRs to countermeasure the process and temperature variations. PTC consumes less energy than DEAP in the range of 28% to 55%, which indicates the energy

efficiency advantage of the nascent photonic PCM devices over conventional ones like MRRs. MERIT further reduces the energy consumption by 12% to 44% compared with PTC as MERIT minimizes the costly programming operations on the photonic PCM devices.

Fig. 12 shows the energy consumption comparison between MERIT and other baseline architectures for mini-batch gradient descent training of six DNN models, assuming each mini-batch includes 128 training examples. One notable change from Fig. 11 is that the reduction in energy consumption of MERIT compared with PTC further increases to 28% to 70%. The computation parallelism and reuse of weight kernels between forward and backward passes of different training examples in a mini-batch are considered the main causes, which cannot be effectively leveraged in stochastic gradient descent training.

C. Breakdown Analysis

1) *Execution Time Breakdown*: Fig. 13 shows the execution time breakdown of the PTC baseline and MERIT architectures when performing mini-batch gradient descent training of the GNMT model, assuming each mini-batch includes 128 training examples. The percentage values are relative to the execution time of the PTC baseline architecture. The overall execution time is divided into three exclusive portions: computation time, time for memory access, and time for programming of photonic PCM devices. Execution time accounts for the cycles in which actual dot product computation is performed. Time for memory access only accounts for the cycles in which no dot product computation is performed and the system is stalled waiting for completion of memory access. PTC baseline architecture takes 40%, 39%, and 22% of overall execution time for computation, memory access, and programming of photonic PCM devices. By contrast, MERIT architecture takes similar time for computation (40%) but less time for memory access (23%) and programming of photonic PCM devices (8%). The main reason behind this is that MERIT incurs minimum programming operations by extensively reusing weights in the forward and backward passes of different training examples within a mini-batch.

2) *Energy Consumption Breakdown*: Fig. 14 illustrates the energy consumption breakdown of PTC baseline and MERIT architectures when performing the mini-batch gradient descent training of GNMT model, assuming each mini-batch includes 128 training examples. The percentage values are relative to the energy consumption of PTC baseline architecture. The overall energy consumption is divided into four exclusive parts: energy related to memory operations, signal transition energy (energy associated with MRRs, TIAs, photodetectors, and DAC/ADC modules), energy related to programming of photonic PCM devices, and energy consumed by the laser source. PTC baseline architecture takes 34%, 46%, 13%, and 7% of overall energy consumption for memory access, signal transition, programming of PCM devices, and laser source. By contrast, MERIT architecture takes less time in all categories. Essentially, memory access, signal transition, and programming of PCM devices are highly related operations. MERIT reuses weights in mini-batch gradient descent training. As a result, a smaller amount of weights will be exchanged between DRAM

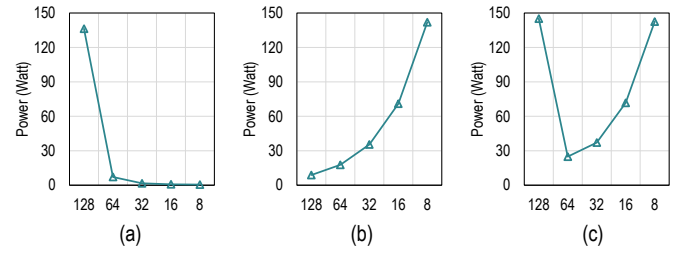


Fig. 15. Evaluation of the impact of photonic PCM array size on (a) laser power, (b) signal transition power, and (c) total power consumption.

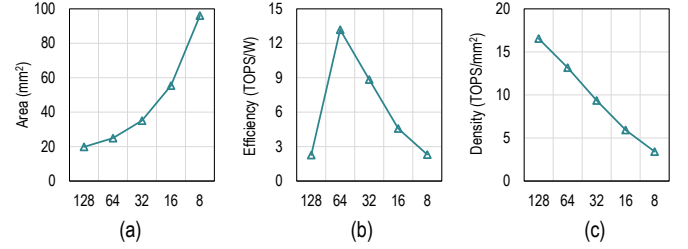


Fig. 16. Evaluation of the impact of photonic PCM array size on (a) chip area, (b) energy efficiency, and (c) computation density.

and SRAM buffer, resulting in less frequent signal transition and programming of photonic PCM devices.

D. Photonic PCM Array Size Trade-off

Fig. 15 demonstrates the evaluation of the impact of photonic PCM array size on the power consumption of the off-chip laser source, the power consumption associated with signal transition between digital and analog domains and electrical and optical domains, and the combined power consumption. We evaluate five different array size configurations: 128, 64 (adopted by this work), 32, 16, and 8 (prototyped by [32]). We also assume 328 TOPS peak performance throughout the evaluation for a fair comparison. For instance, we compare one 128×128 array with four 64×64 arrays throughout the evaluation. We observe that the power consumption of the laser source decreases as the array size decreases, as a smaller array size leads to lower accumulated insertion loss. However, the power consumption associated with signal transition increases, as more DAC/ADC modules and optical components are implemented. Resulting combined effect is shown in Fig. 15 (c) wherein the minimum combined power consumption is reached at an array size of 64.

Fig. 16 shows the evaluation of the influence of photonic PCM array size on area cost, energy efficiency, and computation density. Energy efficiency and computation density are measured in TOPS/W and TOPS/mm², respectively. As the array size decreases, the area cost increases. This is because, though the area for the actual photonic PCM arrays may vary little, area occupied by DAC/ADC modules and other optical components increases. The optimal energy efficiency is reached at an array size of 64 as shown in Fig. 16 (b). Computation density decreases monotonically as the array size decreases as shown in Fig. 16 (c). We adopt 64×64 photonic PCM array

size by collectively considering energy efficiency, computation density, and other factors like quantization loss.

VI. CONCLUSION

We introduce MERIT, a unified photonic DNN accelerator designed to leverage the unique benefits of photonic PCM for high computation parallelism and energy efficiency. MERIT features two main contributions: (1) achieving maximum computation parallelism by leveraging the one-step computation capability of photonic PCM devices and concurrently processing multiple wavelengths and (2) supporting DNN inference and training by mapping weight kernels in photonic PCM arrays and selectively supplying vectors of input features or loss gradients for the dot product computation. Simulation results have proved that MERIT architecture outperforms other state-of-the-art digital and analog DNN accelerators in terms of execution time and energy efficiency.

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